NGHIA VAN VO | Curriculum Vitae

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Summary

COMPUTATIONAL RESEARCHER IN APPLIED MATHEMATICS, OPTIMIZATION, AND MACHINE LEARNING

Extensive experience designing and analyzing optimization algorithms for large-scale ML modeling. Research focuses on translating mathematical optimization into high-performance implementations, including modern fine-tuning and model-compression techniques that make models smaller and more accurate.

Education

OAKLAND UNIVERSITY — 2022–Present
Ph.D. in Applied Mathematics — Rochester, Michigan, USA

OAKLAND UNIVERSITY
M.S. in Applied Statistics — Rochester, Michigan, USA

UNIVERSITY OF EDUCATION — 2017–2021
Mathematics Teacher Education — Ho Chi Minh City, Vietnam

RELEVANT COURSEWORK

Machine Learning; Deep Learning; Advanced Convex Optimization; Nonlinear Programming; Applied Linear Models; Experimental Design; Mathematical Statistics; Time Series; Numerical Methods for PDEs; Numerical Matrix Algorithms.

Publications

1. (with T. A. Nghia Tran, Huy P. N., and V. V. H. Khoa) Lipschitz stability of regularized linear inverse problems, Preprint (2026)

2. (with T. A. Nghia Tran, Huy P. N., and V. V. H. Khoa) Isolated calmness of regularized linear inverse problems, Preprint (2026)

3. (with P. D. Khanh, B. S. Mordukhovich, and D. B. Tran) Double proximal point methods for difference-of-convex optimization methods, Preprint (2026)

4. (with T. A. Nghia Tran and V. V. H. Khoa) An effective subspace Newton method for generalized equations, Preprint (2026)

5. (with P. D. Khanh, V. V. H. Khoa, B. S. Mordukhovich, and D. B. Tran) Inexact DC algorithms in Hilbert spaces with applications to PDE-constrained optimization, arXiv:2601.06622, submitted to **Applied Mathematics & Optimization** (2026)

6. (with H.-C. Luong, Kaiqi Zhao, and L. Chen) Adaptive temperature scaling for knowledge distillation via entropy-regularized supervision, submitted (2026)

7. (with T. A. Nghia Tran and V. V. H. Khoa) Nonsmooth Newton methods with effective subspaces for polyhedral regularization, arXiv:2511.16514, submitted to **Mathematics of Operations Research** (2025)

8. (with T. A. Nghia Tran and Huy P. N.) Stable recovery of regularized linear inverse problems, **Inverse Problems** 41, 065018, DOI: `10.1088/1361-6420/addffb` (2025)

9. (with C.-K. Doan and G.-B. Nguyen) A regularity result via fractional maximal operators for p-Laplace equations in weighted Lorentz spaces, **Complex Variables and Elliptic Equations**, DOI: `10.1080/17476933.2021.1897794` (2022)

10. (with T.-N. Nguyen, M.-P. Tran, and C.-K. Doan) A gradient estimate related fractional maximal operators for a p-Laplace problem in Morrey spaces, **Taiwanese Journal of Mathematics** Advance Publication, 1–21, DOI: `10.11650/tjm/210202` (2021)

Projects

Accelerated Image Processing/Optimization

- Designed and implemented proximal gradient descent, stochastic subgradient, ADMM, and coordinate descent for L1/L2 regression, logistic regression, image denoising, deblurring, and matrix completion.
- Enhanced performance, accuracy, and convergence profiles using Gurobi, scaling experimental throughput to 10x more problem instances.
- Optimized solver parameters, cutting runtime for large-scale problems by up to 60% while maintaining or improving reconstruction quality.

Model Compression using Adaptive Knowledge Distillation

- Developed Adaptive Temperature Scaling (ATS) to teach a small model to mimic a large model for faster and cheaper deployment on resource-constrained platforms, resulting in smaller models with superior accuracy.
- Applied ATS and achieved up to +3.63% accuracy gain over the conventional method without additional training cost, with strong results across multiple CNN architectures including ResNet, VGG, WideResNet, MobileNet, and ShuffleNet on CIFAR-100 and ImageNet data.
- Demonstrated that ATS outperforms SOTA baselines in challenging low-memory settings, keeping tiny models accurate and reliable.

Premier League Ranking Prediction & Disease Symptom Diagnosis

- Conducted EDA and statistical hypothesis testing, including t-tests and ANOVA, to identify high-impact predictive features in structured and unstructured real-world datasets.
- Engineered regression, multi-label classification, and ensemble ML models including XGBoost, SVM, Naive Bayes, KNN, and K-means from scratch in Python with object-oriented design, outperforming Scikit-learn defaults by 3–5% in accuracy.
- Optimized hyperparameters through grid search and cross-validation, improving model generalization to new data.

Teaching Experience

Oakland University — 2022-2027

1. MTH 0661 - Elementary Algebra (Fall 2022, Winter 2023, Winter 2024) - Main Instructor
2. MTH 0662 - Intermediate Algebra (Fall 2023, Winter 2023) - Main Instructor
3. MTH 1441 - Pre-Calculus (Fall 2024) - Main Instructor
4. MTH 2554 - Multivariable Calculus (Fall 2025) - Substitute Instructor
5. MOR 5555 - Nonlinear Optimization (Fall 2025) - Guest Instructor

Honors & Awards

Oakland University — 2025–2026

1. McKay Outstanding Graduate Research Fellowship (2026)
2. Travel Grant to attend Midwest Optimization Meeting (MoM), University of North Dakota (2025), provided by University of North Dakota.
3. Travel Grant to attend East Coast Optimization Meeting (ECOM), George Mason University (2025), provided by George Mason University.
4. Travel Grant to attend Midwest Optimization Meeting (MoM), University of Michigan (2023), provided by University of Michigan.

University of Education, Ho Chi Minh City — 2020–2021

1. The Award for Teaching Excellence Internship, Department of Mathematics, Binh Phu High School (2021)
2. First Prize at Department of Mathematics Undergraduate Scientific Research Competition (2020)
3. First Prize at Undergraduate Scientific Research Competition, H.C.M City University of Education (2020)

Skills

Programming Language Python | Matlab | SAS | C | SQL
Optimization Algorithms Gradient Descent | Stochastic Gradient Descent | Proximal Methods | FISTA | Newton Methods | ADMM | Douglas-Rachford | Chambolle-Pock primal-dual | Coordinate Descent
Frameworks & Tools Scikit-learn | PyTorch | Pandas | NumPy | Matplotlib | Seaborn | CVXPY | Gurobi | CPLEX | MATLAB Optimization Toolbox | Git | GitHub
Certificates Introduction to Data Analytics in Google Cloud | Machine Learning Specialization, Stanford University and Coursera | Scientific Computing and Data Analysis with Python, FreeCodeCamp | Mathematics for Machine Learning, Imperial College London and Coursera
Languages English | Vietnamese